MetaFVG v2: Fixed-Fraction Risk Sizing

Test report, formula derivation, and rollout plan for SL-based strategies

This report documents the switch from static-lot to fixed-fraction risk position sizing introduced in **metafvg_v2**. The new behavior binds trade size to the stop-loss distance so that a stop-out costs exactly **size_position * balance** of account equity. The report covers the formula, the 5-pip FX minimum stop clamp added to protect against near-spread SLs, a regression-test summary, and a proposed rollout to the remaining SL-based strategies with risk fractions calibrated to observed trade frequency.

1. Sizing formula

For every SL-based order, the volume (in MT5 lots) is chosen so that the P&L of hitting the stop-loss exactly equals a fixed fraction of the account balance:

```
volume = (size_position * balance) / (sl_distance * vpu)
```

where **sl_distance = |entry - sl|** (in price units) and **vpu** is the value-per-price-unit (in account currency) of one lot:

```
vpu = contract_size * fx_conv(profit_ccy -> account_ccy)
```

The result is rounded to the symbol's volume_step and clamped to [volume_min, volume_max]. Two safety layers apply on top:

1. 5-pip minimum SL floor (FX only). When trade_calc_mode == 0 (SYMBOL_TRADE_CALC_MODE_FOREX) and the raw SL distance is below 5 pips (0.0001 for non-JPY pairs, 0.01 for JPY), the SL distance used for sizing is raised to 5 pips. The order's actual SL price is unchanged --- only the sizing calculation uses the clamped distance. This protects against near-spread SLs sizing into the maximum-notional cap or into a lot count that inflates fees.

2. 5x-balance notional cap. Regardless of SL distance, the volume is capped so that **volume * contract_size * price (in account ccy) ≤ 5 * balance**. This is a last-resort brake for pathological SL configurations (e.g. a 1-pip SL on an index) --- when it triggers, the trade is placed with strictly less risk than the target, never more.

Failure modes and fallback. If any required MT5 information is missing (account_info, symbol_info, FX conversion rate) or the SL is absent/zero, the function returns **self.size_position** as a raw lot count and logs the reason. This preserves the pre-v2 fixed-lot behavior on failure --- v2 never raises and never blocks a trade.

2. Regression test results

Nine scenarios exercised against mocked MT5 with a $100,000 USD balance and size_position = 0.50%. All pass.

#	Scenario	Price	SL	SL dist	Volume	Real-SL risk	Result
1	EURUSD - normal 20-pip SL	1.1000	1.0980	0.0020	2.50	$500.00	PASS
2	EURUSD - tight 3-pip SL (below floor)	1.1000	1.0997	0.0003	4.55	$136.50	PASS
3	USDJPY - normal 30-pip SL	150.0000	150.3000	0.3000	2.50	$500.00	PASS
4	USDJPY - tight 2-pip SL (below JPY floor)	150.0000	150.0200	0.0200	5.00	$66.67	PASS
5	USDCHF - normal 20-pip SL (non-USD quote)	0.9000	0.8980	0.0020	2.25	$500.00	PASS
6	US500 - 5-point SL	5000.0000	4995.0000	5.0000	1.00	$500.00	PASS
7	US500 - 0.5-point SL (cap only, no pip floor)	5000.0000	4999.5000	0.5000	1.00	$50.00	PASS

8	MTOU - EURUSD 40-pip SL @ 1% risk	1.1000	1.0960	0.0040	2.50	$1,000.00	PASS
9	MLP - GBPCHF 30-pip SL @ 0.25% risk	1.1000	1.0970	0.0030	0.75	$250.00	PASS
10	Fallback - no SL provided (FVGv2)	1.1000	n/a	n/a	0.01	n/a	PASS
11	Fallback - MT5 account_info unavailable	1.1000	1.0980	0.0020	0.01	n/a	PASS

Key observations from the test cases.

Normal SLs (cases 1, 3, 5, 6): realized risk at the real SL is exactly $500 = 0.5% of balance, as designed. The formula lines up across FX with USD quote (EURUSD), JPY (USDJPY, price ~150), non-USD quote (USDCHF), and indices (US500).

Tight-SL FX (cases 2 and 4): when the raw SL is below the 5-pip floor, the sizing uses 5 pips but the trade risk at the *real* SL becomes *smaller* than the target (EURUSD 3-pip SL: $136.50 vs $500 target; USDJPY 2-pip SL: $66.67 vs $500 target). This is deliberate --- the alternative was to size against the tiny raw distance and hit the notional cap at a lot count that would make broker fees and slippage dominate.

Very-tight index SL (case 7): indices bypass the 5-pip clamp (calc_mode ≠ 0), so the notional cap catches this: 0.5-point SL on US500 would raw-size to 10 lots, the cap trims to 1 lot, realized risk is $50. Same principle --- trade underrisks, never overrisks, when the SL geometry is pathological.

Fallback (cases 8, 9): when SL is absent or MT5 info is unavailable, the raw *size_position* is returned as-is. Under the new interpretation of *size_position* as a fraction (0.005), this means the fallback lot size is 0.005 lots --- effectively skipping the trade. This is the intended behavior for v2, where a missing SL is a bug that should not silently sail through as a full position size.

3. SL-based strategy activity (last 180 days)

Live MT5 deal counts, 2026-02-06 to 2026-08-05. Only open deals (entry == 0) are counted. MetaFVG v1 dominates by two orders of magnitude vs everything else; that fact drives the per-strategy risk-fraction recommendation in section 4.

Strategy	SL-based?	Total trades	Trades/month	Tags	Trades/mo per tag
metafvg	yes	4,339	733.8	4	183.44
metafvg_v2	yes	8	1.4	4	0.34
metamlp	yes	659	111.4	7	15.92
metaob	yes	48	8.1	10	0.81
metamtou	yes	10	1.7	7	0.24
metane	no	1,563	264.3	3	88.11
metaga	no	500	84.6	4	21.14
metago	no	145	24.5	15	1.63

Rows in muted colour are market-order strategies without SLs --- listed for context only, they are not affected by this change.

4. Recommended risk fraction per SL strategy

Sizing rule of thumb: keep expected monthly loss volatility of each strategy at ~2% of balance when trades are treated as independent draws. Under a Bernoulli model with per-trade risk r, the monthly loss standard deviation across N trades scales as **r · sqrt(N)**. Setting that to 0.02 gives a soft upper bound **r_max** ≈ **0.02 / sqrt(N)**. Real trades are correlated (multiple instances of the same strategy fire on overlapping regimes), so the recommendations below apply a further ~2x safety margin on top of that.

Strategy	Trades/mo	r_max soft	Current	Recommended	Reason
metafvg	733.8/mo	0.07%	0.01--0.03 lot	0.05%	very high freq: 4 tags & 730+ trades/mo. 0.05% keeps monthly lo
metafvg_v2	1.4/mo	1.72%	0.50%	0.50%	already fractional; live sample small but backtested at 0.5%.
metamlp	111.4/mo	0.19%	0.05 lot	0.25%	medium-high freq (110/mo across 7 tags). 0.25% ~ 0.5% monthly
metaob	8.1/mo	0.70%	0.01--1.0 lot	0.30%	wide config range today; low freq (8/mo). 0.30% risk uniformly res
metamtou	1.7/mo	1.54%	0.01 lot	0.50%	very low freq (1.7/mo); daily bars, high conviction. 0.50% matche

What changes on the strategy classes. Each SL-based strategy needs the same one-method override that *metafvg_v2* already has --- a *_resolve_position_size* that (a) reinterprets *size_position* from the YAML as a fraction of balance, (b) computes lots from balance / SL distance / value-per-price-unit, (c) applies the 5-pip FX minimum and 5x notional cap, (d) falls back to the raw *size_position* on any MT5 error. The math is identical to v2 --- only the interpretation of *size_position* in the YAML config changes.

MetaScale interaction. All strategies in *MetaScale.DEFAULT_STRATEGY_PARAMS* (metafvg, metago, metane, metaga, metaob) have their *size_position* periodically overwritten by MetaScale's CVXPY optimizer. Switching those strategies to fixed-fraction interpretation means MetaScale would need to be aware of that change --- either write fractions (and its covariance model reinterpreted accordingly) or those strategies drop out of MetaScale's book like *metafvg_v2* already does. This is the main open decision before rolling out the change to *metafvg* v1 and *metaob*.

5. Suggested rollout order

Phase 1 (low-risk, no MetaScale conflict). Add *_resolve_position_size* override + YAML rewrite for **metamtou** and **metamlp**. Neither is currently in MetaScale's *DEFAULT_STRATEGY_PARAMS*, so this can ship without touching that logic. MTOU trades are infrequent and daily-bar, so a bad week is bounded; MLP trades at medium-high frequency, so this is where the sizing improvement will actually compound.

Phase 2 (requires MetaScale change). Extend to **metaob** (and later **metafvg** v1). This needs a corresponding update to *MetaScale*: either remove these strategies from *DEFAULT_STRATEGY_PARAMS* (letting them manage their own sizing like v2), or teach *MetaScale* to write fractions in the fractional-sizing convention and compute portfolio covariance against value-per-price-unit rather than tick_value.

Verification per strategy. Before flipping any YAML, run the same mocked-MT5 unit test used for v2 (this report, section 2) against each strategy's new *_resolve_position_size*. Then dry-run each strategy under PM2 with **debug=True** for a few LTF bars and inspect the sizing log lines (*balance*=... *risk*=... *sl_dist*=... -> *volume*=...) before restarting for live production.

Appendix A: Sample log output

Verbatim log lines captured by the mocked test runner for scenario 1 (EURUSD 20-pip SL) --- the format shipped in production is identical.

```
EURUSD: balance=100000.00 USD risk=0.50% -> risk_amount=500.00 sl_dist=0.002000 sl%=0.182%
vpu=100000.0000 raw=2.5000 -> volume=2.5
```

Appendix B: Scenario 2 log (5-pip clamp fires)

```
EURUSD: SL distance 0.000300 below 5-pip minimum (0.000500), clamping for sizing
```

```
EURUSD: notional cap triggered (1100000 > 500000) -> volume 10.0 -> 4.55
```

```
EURUSD: balance=100000.00 USD risk=0.50% -> risk_amount=500.00 sl_dist=0.000500 sl%=0.045%
vpu=100000.0000 raw=10.0000 -> volume=4.55
```

Appendix C: Scenario 7 log (notional cap fires on index)

```
US500: notional cap triggered (5000000 > 500000) -> volume 10.0 -> 1.0
```

```
US500: balance=100000.00 USD risk=0.50% -> risk_amount=500.00 sl_dist=0.500000 sl%=0.010%
vpu=100.0000 raw=10.0000 -> volume=1.0
```